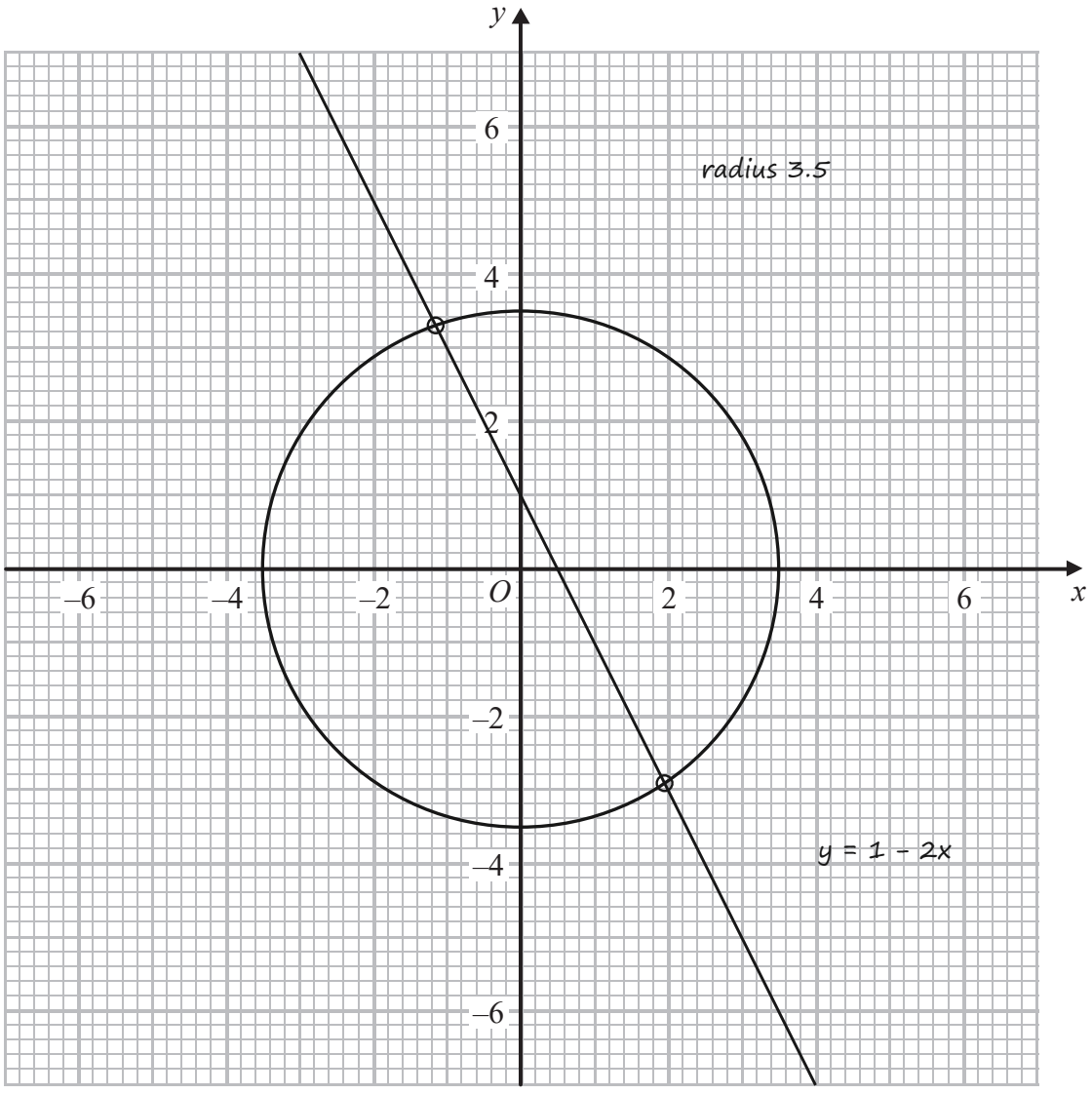
16 (a) On the grid, draw the graph of $x^2 + y^2 = 12.25$



2 marks total
Circle centre (0, 0), radius √12.25 = 3.5

(2)

(b) Hence find estimates for the solutions of the simultaneous equations

$$x^2 + y^2 = 12.25$$
$$2x + y = 1$$

Draw y = 1 – 2x through (0, 1) and (1, –1).

1 mark
Use the straight line to find intersections

(2.0, –2.9), (–1.2, 3.3)

1 mark
One correct matched pair

1 mark
Both pairs, matched clearly

(3)

(Total for Question 16 is 5 marks)